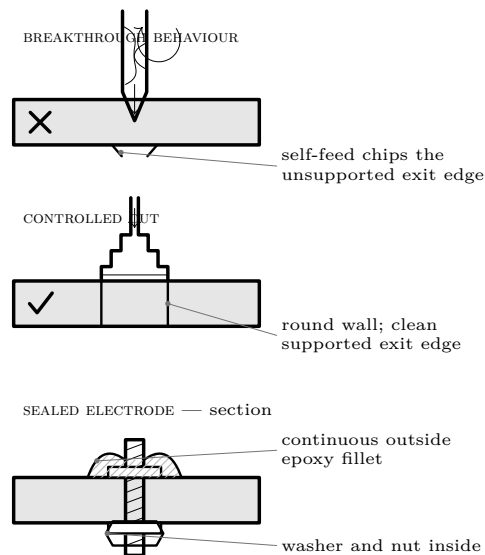


# Craft 5: Drill & Seal

Every hole you make is a hole in a pressure vessel

## THE STANDARD

A round hole of the intended size, with no cracks radiating from it, no crazing around it, and a seal that is continuous all the way round on the **outside** face. Nothing you fit through a chamber wall may rely on the threads alone to hold pressure.



### PVC grabs

A standard twist drill's cutting geometry pulls itself into plastic. As the point breaks through, the flutes snatch, and the drill either climbs through violently or cracks the far wall. This surprises everyone once.

### Use a step drill

It cannot self-feed, it leaves a clean round hole, and it works from a small pilot upward. This is the right tool and it is inexpensive.

### Or grind the drill

If you only have twist drills, back the cutting edge off to near zero rake, or drill a pilot and then open up in stages. Slow speed, light steady pressure, and back off as you approach breakthrough.

### Back it up

Clamp a scrap block behind the wall where you can. It supports the material at the moment of breakthrough, which is the moment things go wrong.

### Never near a joint

Keep penetrations well away from a welded socket. The wall there is a different thickness and the weld zone has different properties.

### Round holes only

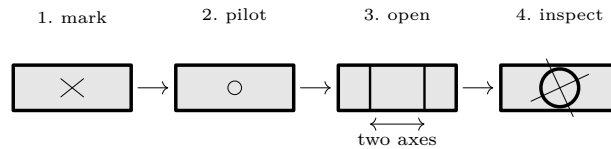
No slots, no filed-out ovals, no drilling two overlapping holes. Every corner is a stress concentration and a crack starter.

## Method for the igniter electrodes

1. **Mark.** Put each centre cross at the drawing dimensions, clear of every welded joint. *see the **Craft 1: Measure & Mark** sheet*
2. **Label.** Write the specified diameter beside each cross.
3. **Measure.** Record each centre from the drawing datum.
4. **Decide.** If either measurement differs from the drawing, erase and remark before cutting.
5. **Pilot.** Drill one small hole at low speed with backing behind the exit face.
6. **Verify.** Measure the pilot centre from the same datum. An off-centre pilot requires a new part; never steer, slot or overlap it.
7. **Open.** Use a step drill to reach the specified diameter, reducing pressure before breakthrough.
8. **Measure.** Take two finished diameters at 90°.
9. **Decide.** Both readings must match the drawing and each other. An oval or oversize hole requires a new part.
10. **Deburr.** Remove the edge burr on both faces. *see the **Craft 3: Deburr & Chamfer** sheet*
11. **Inspect.** Rotate the hole under raking light. Any crack, white line or craze requires a new part.
12. **Fit.** Install the screw, one washer on each face and the inside nut; tighten only enough to hold position.
13. **Set.** Adjust the internal electrode gap to 3–5 mm.
14. **Test.** With the chamber open, use a battery-free piezo click and confirm one visible, audible spark. No spark means adjust and repeat before sealing.
15. **Seal.** Form one continuous outside epoxy fillet around the screw.
16. **Inspect.** Rotate the part and view the complete fillet. A gap, bubble or lifted edge means remove it before cure and reseal.

17. **Record.** Write the epoxy cure start date and time.

18. **Wait.** Do not handle or load the penetration until the manufacturer's full cure time has elapsed.



#### FOUR STOPS BEFORE SEALANT

**After marking:** Is the centre referenced to a datum, not guessed? **After the pilot:** Is it still on the cross? **After opening:** Do two diameter readings agree? **After deburring:** Does raking light show a clean, round edge with no white lines? Record each answer. A doubtful answer means remake, not repair.

#### TEST BEFORE YOU SEAL

Once the epoxy is on, changing the gap means destroying the seal. Test the spark with a battery-free piezo click while you can still reach both electrodes, and confirm you see and hear it jump. Sealing an untested igniter is how a build ends up needing a new chamber.

### Why a crack near a hole is different

A crack in plain pipe wall has to start somewhere. A hole gives it a start: the stress in the material flows around the hole and concentrates at its edge, roughly tripling there even for a perfectly round hole. A sharp corner, a scratch from a slipped drill, or a star crack from a snatching bit multiplies that far further. This is why the rules are round holes, no sharp corners, no scratches, and inspect every penetration before every session.

**Check it yourself.** Hold the drilled area up to a strong light and look at the plastic around the hole from several angles. You are looking for **crazing** — a faint web of tiny white lines — and for any crack, however short. Crazing means the material was stressed while you drilled it. Either is a reason to remake the part, not to seal over it.

#### The habit worth taking away

Match the tool to the material rather than forcing the tool you have. A twist drill is a superb tool for steel and a poor one for thin plastic, and no amount of care fully compensates. Recognising that a job wants a different tool — and that the right one often costs very little — is most of what separates work that looks made from work that looks got away with.

#### Final proof

Before accepting the part, make one record containing the drawing diameter, both measured diameters, distance from the chosen datum, cure start time and cure-complete time. Add one close photograph in raking light before sealant and one after cure, with the complete fillet visible. Initial the record. This is proof that the checks happened; it is not permission to test with pressure.

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